

DEEP LUKHI

Aspiring Software Engineer — C++ & Data Structures

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Summary

Software engineering undergraduate with a strong foundation in **C++ and Data Structures and Algorithms**. Experience building **command-line applications** and **graph-based systems**. Solved **550+ algorithmic problems**, focusing on clean code, correctness, and efficiency.

Skills

Programming Languages: C, C++

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, STL

Developer Tools: Git, GitHub, VS Code, Linux (basic)

Problem Solving Platforms: LeetCode, GeeksforGeeks, CodeStudio

Projects

CLI Authentication System | C++, OOP, File Handling, Security

- Built a secure, modular C++ CLI authentication system with registration, login, and password change workflows.
- Implemented password hashing, input validation, and file-based persistence with per-user credential files.
- Designed clean layered architecture separating UI, business logic, validation, and utilities.
- Added unit tests for validators and automated builds using Makefile.

Graph-Based Movie Recommendation System | C++, Graphs, DSA

- Developed a graph-based recommendation engine modeling users and items as nodes with interactions as edges.
- Implemented multiple algorithms including Collaborative Filtering, Popularity-Based, and BFS-based recommendations.
- Used adjacency lists for efficient graph traversal with time complexity up to $O(V + E)$.
- Enabled persistent storage, modular algorithm extension, and automated builds via Makefile.

Flight Route Planner | C++, Graph Algorithms, Dijkstra

- Built a professional-grade flight routing system modeling cities and flights as weighted graphs.
- Implemented Dijkstra's Algorithm to compute both cheapest (cost) and fastest (time) routes.
- Added file-based persistence for routes, color-coded CLI interface, and comprehensive test suite.
- Achieved optimal routing with time complexity $O((V + E) \log V)$ using priority queues.

Achievements & Activities

- **Hackathon Participant:** Participated in multiple intra-college and inter-college hackathons, demonstrating rapid prototyping abilities, collaborative problem-solving, and delivery of functional solutions under time constraints.
- **Competitive Programming:** Solved **550+ Data Structures and Algorithms problems** across **LeetCode**, **GeeksforGeeks**, and **CodeStudio**, with consistent focus on optimization and accuracy.
- **GeeksforGeeks University Rank:** Achieved **Rank 1** on GeeksforGeeks at the university level, reflecting strong problem-solving skills and consistent competitive performance.

Education

Bachelor of Technology in Information Technology

Bhagwan Mahavir University, Surat

2023 – 2027

CGPA: 8.1 / 10

Additional Information

Key Strengths: Strong analytical and problem-solving abilities, quick learner with adaptability to new technologies, effective team collaboration, and commitment to writing clean, scalable, and maintainable code.